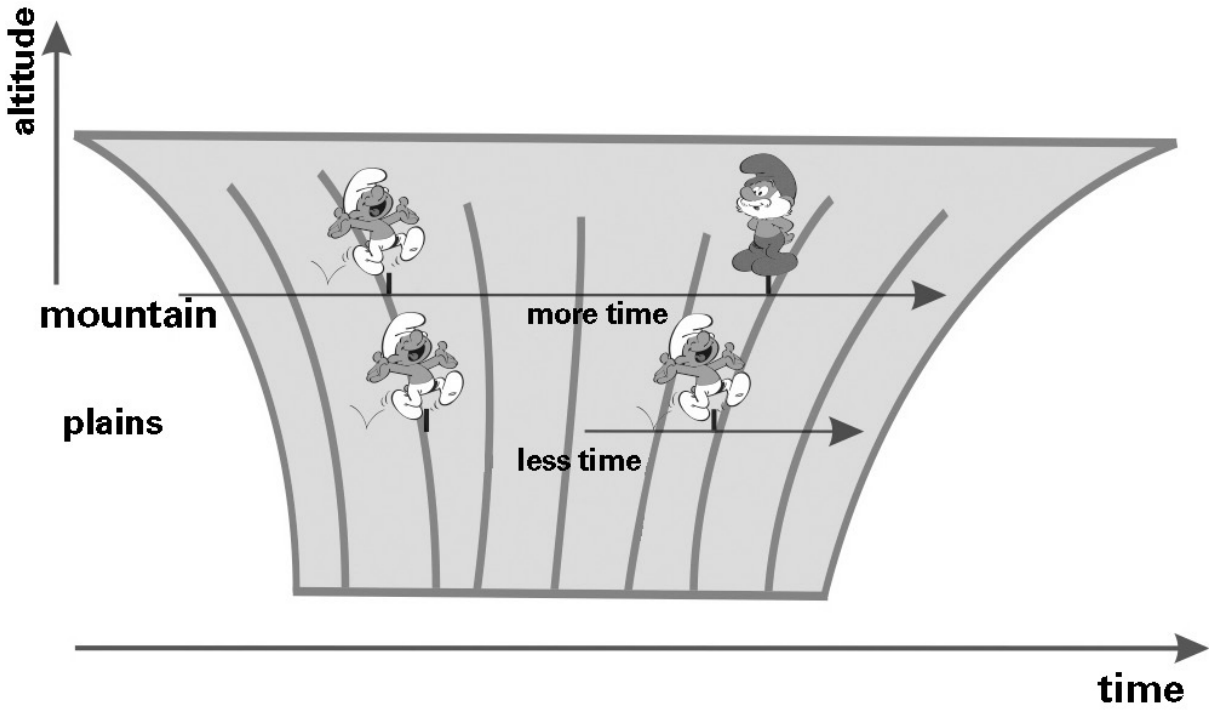


The image above illustrates what physicists call “curved” spacetime. “Curved” because it is distorted: distances are stretched and contracted, just like the elastic sheet when it is pulled. This is why the light cones were inclined in the diagrams in chapter 3.

Time thus becomes part of a complicated geometry woven together with the geometry of space. This is the synthesis that Einstein found between Aristotle’s conception of time and Newton’s. With a tremendous beat of his wings, Einstein understands that Aristotle and Newton are *both* right. Newton is right in intuiting that something else exists in addition to the simple things that we see moving and changing. True and mathematical Newtonian time exists; it is a real entity; it is the gravitational field, the elastic sheet, the curved spacetime in the diagram. But Newton is wrong in assuming that this time is independent from things—and that it passes regularly, imperturbably, separately, from everything else.

For his part, Aristotle is right to say that “when” and “where” are always located in relation to something. But this something can also be just the field, the spatiotemporal entity of Einstein. Because this is a dynamic and concrete entity, like all those in reference to which, as Aristotle rightly observed, we are capable of locating ourselves.

All this is perfectly coherent, and Einstein’s equations describing the distortions of the gravitational field and its effects on clocks and meters have been repeatedly verified for more than a century. But our idea of time has lost